Def. Let $f:(a,b)\to\mathbb{R}$ be a map, and $x\in[a,b]$.
If the limit exists, define

$$f'(x) \stackrel{\text{def}}{=} \lim_{t\to x}\frac{f(t)-f(x)}{t-x} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

This $f'(x)$ is called the derivative of f at x.

More rigorously, given $\varepsilon>0$, there exists $\delta>0$ s.t

$$0<|t-x|<\delta \Rightarrow \left|\frac{f(t)-f(x)}{t-x}-f'(x)\right|<\varepsilon \quad \text{or} \quad 0<|h|<\delta \Rightarrow \left|\frac{f(x+h)-f(x)}{h}-f'(x)\right|<\varepsilon$$

Note: letting $h=t-x$, these are same. Sometimes, we can use the following equivalent:

Lemma. Let $f:(a,b)\to\mathbb{R}$ be a map, and $x\in[a,b]$. TFAE:

a) $f'(x)$ exists.

b) There exists a map $\eta: B_\delta(0)\setminus\{0\}\to\mathbb{R}$ for some $\delta>0$ and $L\in\mathbb{R}$ s.t

$$f(x+h)-f(x) = L\cdot h + |h|\cdot\eta(h) \quad \text{and} \quad \eta(h)\to 0 \text{ as } h\to 0.$$

Proof. a) $\Rightarrow$ b) Given $\varepsilon>0$, there exists $\delta>0$ s.t

$$0<|h|<\delta \Rightarrow \left|\frac{f(x+h)-f(x)}{h}-f'(x)\right|<\varepsilon.$$

Now, let $L=f'(x)$ and define for each $0<|h|<\delta$

$$\eta(h) = \frac{1}{|h|}\cdot\left(f(x+h)-f(x)-L\cdot h\right)$$

Then $\lim_{h\to 0}\eta(h)=0$. Conversely, if such η exists, then

$$\frac{f(x+h)-f(x)}{h} = L + \frac{|h|}{h}\cdot\eta(h)\to L \text{ as } h\to 0,$$

therefore a) is satisfied

Note: the absolute value above η is <u>Not</u> necessary in this case,
But it is <u>necessary</u> when our ambient space is <u>Not</u> $\mathbb{R}^1$, is $\mathbb{R}^p$.

Thm. Let $f, g : (a,b) \to \mathbb{R}$ be maps, and let $x \in (a,b)$.
If f, g are differentiable at x, then $f+g$, $f \cdot g$, f/g are differentiable at x with
1) $(f+g)'(x) = f'(x) + g'(x)$
2) $(f \cdot g)'(x) = f'(x)g(x) + f(x)g'(x)$
3) $(f/g)'(x) = (f'(x)g(x) - f(x)g'(x))/g(x)^2$

Proof. Given $\varepsilon > 0$, there exists $\delta > 0$ s.t.

$$\left| \frac{f(x+h) - f(x)}{h} - f'(x) \right|, \left| \frac{g(x+h) - g(x)}{h} - g'(x) \right| < \frac{\varepsilon}{2} \quad \text{if } 0 < |h| < \delta.$$

This implies

$$\left| \frac{f(x+h) + g(x+h) - (f(x) + g(x))}{h} - (f'(x) + g'(x)) \right| < \varepsilon$$

By the triangle inequality. To show 2), observe that

$$f(x+h)g(x+h) - f(x)g(x) = f(x+h) \cdot (g(x+h) - g(x)) + g(x) \cdot (f(x+h) - f(x))$$

Dividing both side by h, and letting $h \to 0$. Then we obtain the result.
Similarly, observe that

$$\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} = \frac{f(x+h)g(x) - f(x)g(x+h)}{g(x+h) \cdot g(x)}$$

$$= \frac{1}{g(x+h) \cdot g(x)} \cdot \left[(f(x+h) - f(x)) \cdot g(x) - f(x)(g(x+h) - g(x)) \right]$$

Dividing ... Yeah!

Thm. Let $f:(a,b)\to\mathbb{R}$ and $g:I\to\mathbb{R}$ be functions, where $\mathrm{Im}\, f\subseteq I$, and let $x\in(a,b)$. If $f'(x)$ exists and $g'(f(x))$ exists, then $g\circ f$ is differentiable at x with

$$(g\circ f)'(x) = g'(f(x))\cdot f'(x)$$

Proof. Using the equivalent condition for differentiable: there exists η_1, η_2 s.t

$$f(x+h_1)-f(x) = h_1\cdot f'(x) + |h_1|\cdot \eta_1(h_1) \quad \text{and} \quad \lim_{h_1\to 0}\eta_1(h_1)=0$$

$$g(f(x)+h_2)-g(f(x)) = h_2\cdot g'(f(x)) + |h_2|\cdot\eta_2(h_2) \quad \text{and} \quad \lim_{h_2\to 0}\eta_2(h_2)=0.$$

Now, let $h_2 = f(x+h_1)-f(x)$ then (It is possible since $f(x+h_1)-f(x)\to 0$ as $h_1\to 0$

$$(g\circ f)(x+h_1)-(g\circ f)(x) = g(f(x)+h_2)-g(f(x)) = h_2\, g'(f(x)) + |h_2|\cdot \eta_2(h_2)$$

Dividing by h_1 and letting $h_1\to 0$ then

$$\frac{(g\circ f)(x+h_1)-(g\circ f)(x)}{h_1} = \frac{h_2}{h_1}\cdot g'(f(x)) + \frac{|h_2|}{h_1}\cdot\eta_2(h_2) \to f'(x)\cdot g'(f(x))$$

Problem. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable with $\sup |f'(x)| < 1$.
Then f has a fixed point.

Proof. Construct a sequence. Let $x_1 = \alpha \in \mathbb{R}$, $x_{n+1} = f(x_n)$ $n \geq 1$.
Take $\beta = \sup |f'(x)| < 1$. Given $n \in \mathbb{N}$, if $x_n = x_{n-1}$ then x_{n-1} is the fixed point. If $x_n = x_{n-1}$,

$$|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| = |x_n - x_{n-1}| \cdot |f'(c)| \leq \beta \cdot |x_n - x_{n-1}| \quad \text{by the MVT.}$$

Now,

$$\begin{aligned} |x_{n+1} - x_n| &= |f(x_n) - f(x_{n-1})| \leq \beta |x_n - x_{n-1}| \\ &= \beta \cdot |f(x_{n-1}) - f(x_{n-2})| \leq \beta^2 |x_{n-1} - x_{n-2}| \\ &\ \ \vdots \\ &\leq \beta^{n-1} |x_2 - x_1|. \end{aligned}$$

Given $\varepsilon > 0$, put $N \in \mathbb{N}$ such that $\beta^{N-1} |x_2 - x_1| < \varepsilon (1 - \beta)$
Then for any $m \geq n \geq N$,

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} + x_{m-2}| + \cdots + |x_{n+1} - x_n| \\ &\leq \left(\beta^{m-2} + \beta^{m-1} + \cdots + \beta^{n-1}\right) \cdot |x_2 - x_1| \\ &= \beta^{n-1} \left(1 + \beta + \cdots + \beta^{m-n-1}\right) \cdot |x_2 - x_1| \\ &< \beta^{n-1} \left(\sum_{k=0}^{\infty} \beta^n\right) \cdot |x_2 - x_1| = \beta^{n-1} \cdot \frac{1}{1 - \beta} \cdot |x_2 - x_1| < \varepsilon. \end{aligned}$$

Therefore $\{x_n\}$ is cauchy sequence in $\mathbb{R}$, the complete metric space, thus

$$x^* = \lim_{n \to \infty} x_n$$

exists. Finally, the differentiable map continuous,

$$f(x^*) = f\left(\lim_{n \to \infty} x_n\right) = \lim_{n \to \infty} f(x_n) = \lim_{n \to \infty} x_{n-1} = x^*. \qquad \blacksquare$$